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20 / 20 submitted

Q1. Union of Sets

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {3, 4, 5}
c = A.union(B)
print("{ " + ",".join(map(str,sorted(c))) + " ", end=" ")
```

OUTPUT

{1,2,3,4,5}

Q2. Intersection

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1,2,3}
B = {2,3,4}
C = A.intersection(B)
print("{ " + ",".join(map(str,sorted(C))) + " ", end=" ")
```

OUTPUT

{2,3}

Q3. Difference

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1,2}
B = {2,3,4}
C = sorted(A^B)
s = ""
for i in C:
    s = s + str(i) + ","
print("{ " + s[:1] + "}")
```

OUTPUT

{1}

Q4. Symmetric Difference

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1,2,3}
B = {2,3,4}
C = sorted(A^B)
if len(C) == 0:
    print("{}")
else:
    S = ""
    for i in C:
        S = S + str(i) + ","
    print("{ " + S[:-1] + "}")
```

OUTPUT

{1,4}

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Q5. Subset

Score: 5.00 TC: 1/1 passed

CODE

```
A = {2,3}
B = {1,2,3,4}
if A.issubset(B):
    print("Subset")
else:
    print("Not Subset")
```

OUTPUT

Subset

Q6. Superset

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1,2,3,4}
B = {2,3}
if A.issuperset(B):
    print("Superset")
else:
    print("Not Superset")
```

OUTPUT

Superset

Q7. Disjoint Sets

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1,2}
B = {3,4}
if A.isdisjoint(B):
    print("Disjoint")
else:
    print("Not Disjount")
```

OUTPUT

Disjoint

Q8. Remove Duplicates

Score: 5.00 TC: 1/1 passed

CODE

```
lst = [1,2,2,3,3,4]
s = set(lst)
u = sorted(s)
res = ""
for i in u:
    res = res + str(i) + ","
print("{ " + res[:-1] + "}")
```

OUTPUT

{1,2,3,4}

Q9. Common Elements

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1,2,3}
B = {2,3,4}
C = {2,5}
common = sorted(A & B & C)
if len(common) == 0:
    print("{}")
else:
    s = ""
    for i in common:
        s = s + str(i) + ","
    print "{" + s[:-1] + "}"
```

OUTPUT

{2}

Q10. Missing Numbers

Score: 5.00 TC: 1/1 passed

CODE

```
Given = {1,2,4,5,7,10}
full = set(range(1, 11))
missing = sorted(full - Given)
if len(missing) == 0:
    print("{}")
else:
    s = ""
    for i in missing:
        s = s + str(i) + ","
    print "{" + s[:-1] + "}"
```

OUTPUT

{3,6,8,9}

Q11. First Repeated

Score: 5.00 TC: 1/1 passed

CODE

```
lst = input().split()
seen = set()
for x in lst:
    num = int(x)
    if num in seen:
        print(num)
        break
    else:
        seen.add(num)
```

INPUT

10 20 30 20 40

OUTPUT

20

Q12. Duplicate Elements

Score: 5.00 TC: 1/1 passed

CODE

```

lst = input().split()
seen = set()
duplicates = set()
for x in lst:
    num = int(x)
    if num in seen:
        duplicates.add(num)
    else:
        seen.add(num)
dup_list = sorted(list(duplicates))
s = '{' + ','.join(str(i) for i in dup_list) + '}'
print(s)

```

INPUT 10 20 30 20 10**OUTPUT** {10,20}**Q13. Compare Lists**

Score: 0.00 TC: 0/1 passed

CODE

```

list1 = input()
list2 = input()

s1 = s1.replace('list1','').replace(':', '').strip()
s2 = s2.replace('list2','').replace(':', '').strip()

list1 = s1.split()
list2 = s2.split()

set1 = []
set2 = []

for x in list1:
    if x not in set1:
        set1.append(x)

for x in list2:
    if x not in set2:
        set2.append(x)

set1.sort()
set2.sort()

if set1 == set2:
    print(True)
else:
    print(False)

```

INPUT list1 :1 2 2 3
list2 :3 2 1**OUTPUT** File "C:\Windows\TEMP\sandbox_192421390RD_6a722f0d10bec8.59158971\main.py", line 4
s1 = s1.replace('list1','').replace(':', '').strip()
SyntaxError: unmatched ')'

Q14. Duplicate Tuples

Score: 0.00 TC: 0/1 passed

CODE

```
s = input()
s = s.replace('[', '').replace(']', '')
lst = s.split(',')
res = []
for item in lst:
    item = item.replace('(', '').replace(')', '')
    x,y = item.split(',')
    pair = (int(x), int(y))
    if pair not in res:
        res.append(pair)
print(res)
```

INPUT [(1,2),(1,2),(3,4)]

OUTPUT [(1, 2), (3, 4)]

Q15. Pair Sum

Score: 0.00 TC: 0/1 passed

CODE

```
a = list(map(int, input().split()))
t = int(input())
res = []
for i in range(len(a)):
    for j in range(i+1, len(a)):
        if a[i] + a[j] == t:
            res.append(f"({a[i]}, {a[j]})")
print(" ".join(res))
```

INPUT 2 4 6 8 10
12

OUTPUT (2,10),(4,8)

Q16. Unique Characters

Score: 5.00 TC: 1/1 passed

CODE

```
s = input()
print(len(set(s)))
```

INPUT programming

OUTPUT 8

Q17. Unique Words

Score: 0.00 TC: 0/1 passed

CODE

```
s = input()
words = s.split()
unique = set(words)
s = '{ ' + ', '.join(repr(i) for i in sorted(unique)) + '}'
print(s)
```

INPUT python is easy python

OUTPUT {'easy', 'is', 'python'}

Q18. Exactly One Set

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1,2,3}
B = {3,4,5}
C = {5,6,7}
res = (A - B - C) | (B - A - C) | (C - A - B)
print("{ " + ",".join(map(str, sorted(res))) + "}")
```

OUTPUT

{1,2,4,6,7}

Q19. Menu Driven**CODE**

```
A = set(map(int, input().replace(' ', '').split()))
B = set(map(int, input().replace(' ', '').split()))
while True:
    c = int(input())
    if c == 1: print("Union:", sorted(A | B))
    if c == 2: print("Intersection:", sorted(A & B))
    if c == 3: print("Diffence:", sorted(A - B))
    if c == 4: print("Symmetric Difference:", sorted(A ^ B))
    if c == 5: break
```

INPUT

```
1,2,3,4
3,4,5,6
1
5
```

OUTPUT

Union: [1234, 3456]

Q20. Student Database**CODE**

```
s = set()
while True:
    try:
        C = int(input())
    except:
        break
    if C == 1:
        n = input()
        s.add(n)
        print("Student added successfully.")
    elif C == 2:
        if len(s) == 0:
            print("Not students in database.")
        else:
            for name in sorted(s):
                print(name)
    elif c == 3:
        break
```

INPUT

```
1
Ravi
```

OUTPUT

Student added successfully.